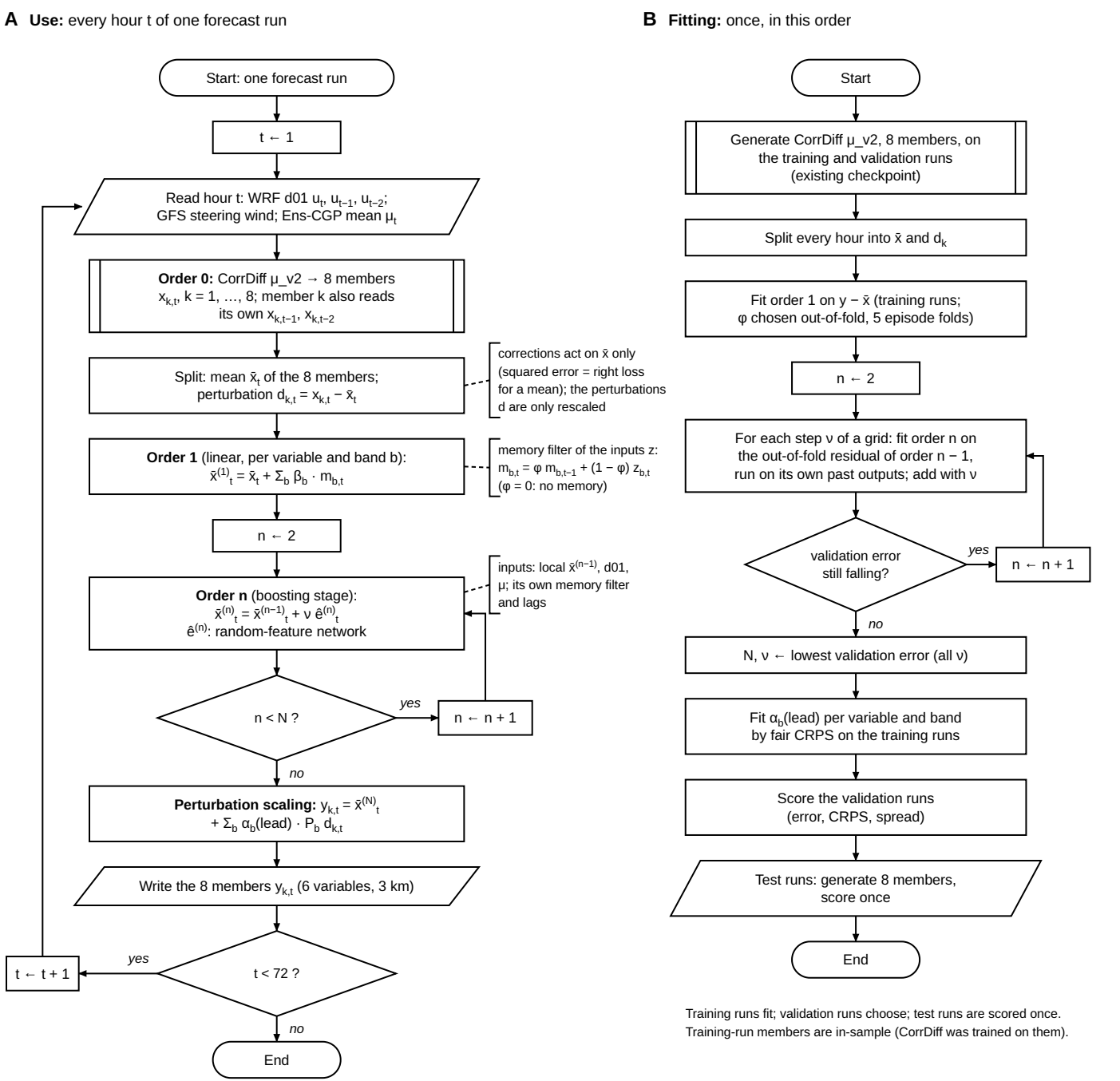

A Use: every hour t of one forecast run
Start: one forecast run
t ← 1
Read hour t: WRF d01 u_t, u_t−1, u_t−2; GFS steering wind; Ens-CGP mean μ_t
Order 0: CorrDiff μ_v2 → 8 members x_k,t, k = 1, …, 8; member k also reads its own x_k,t−1, x_k,t−2
Split: mean x̄_t of the 8 members; perturbation d_k,t = x_k,t − x̄_t
corrections act on x̄ only (squared error = right loss for a mean); the perturbations d are only rescaled
Order 1 (linear, per variable and band b): x̄(1)_t = x̄_t + Σ_b β_b · m_b,t
memory filter of the inputs z: m_b,t = φ m_b,t−1 + (1 − φ) z_b,t (φ = 0: no memory)
n ← 2
Order n (boosting stage): x̄(n)_t = x̄(n−1)_t + ν ê(n)_t ê(n): random-feature network
inputs: local x̄(n−1), d01, μ; its own memory filter and lags
n < N ?
yes
n ← n + 1
no
Perturbation scaling: y_k,t = x̄(N)_t + Σ_b α_b(lead) · P_b d_k,t
Write the 8 members y_k,t (6 variables, 3 km)
t < 72 ?
yes
t ← t + 1
no
End
B Fitting: once, in this order
Start
Generate CorrDiff μ_v2, 8 members, on the training and validation runs (existing checkpoint)
Split every hour into x̄ and d_k
Fit order 1 on y − x̄ (training runs; φ chosen out-of-fold, 5 episode folds)
n ← 2
For each step ν of a grid: fit order n on the out-of-fold residual of order n − 1, run on its own past outputs; add with ν
validation error still falling?
yes
n ← n + 1
no
N, ν ← lowest validation error (all ν)
Fit α_b(lead) per variable and band by fair CRPS on the training runs
Score the validation runs (error, CRPS, spread)
Test runs: generate 8 members, score once
End
Training runs fit; validation runs choose; test runs are scored once.
Training-run members are in-sample (CorrDiff was trained on them).
Symbols:
start / end
process
model run as one block
input / output
decision
note